DDR SDRAM - TECHNICAL REPORT

CORE WORKING PRINCIPLE

Double Data Rate Mechanism

DDR SDRAM transfers data on **both rising and falling edges** of the clock signal, doubling data throughput without increasing clock frequency. While SDRAM transfers data only on the rising edge (single data rate), DDR exploits both transitions.

Data Rate Calculation:

- Clock frequency × 2 (both edges) × Bus width (64 bits) ÷ 8 = Transfer rate
- Example: 100 MHz clock → 200 MT/s effective → 1600 MB/s peak transfer

2n Prefetch Architecture (DDR1)

The fundamental innovation enabling DDR is the **prefetch buffer**:

Internal Architecture:

- Internal memory core operates at **single clock edge** (like SDRAM)
- Internal data bus width = **2× external I/O bus width**
- Memory core fetches 2 data words simultaneously into prefetch buffer
- I/O buffer then outputs these words on consecutive clock edges

How it Works:

1. Memory core reads 2n bits from cell array in one core clock cycle
2. Data placed in I/O buffer as two separate words
3. First word transmitted on rising edge
4. Second word transmitted on falling edge
5. This converts slow single-edge core operation to fast double-edge I/O

Example (16-bit wide chip):

- Core fetches 32 bits (2×16) per access
- I/O releases 16 bits on rising edge, 16 bits on falling edge
- Effective I/O speed = 2× core speed

CLOCK ARCHITECTURE

Differential Clock (CK/CK#)

Unlike SDRAM's single-ended clock, DDR uses differential clocking:

- Two complementary clock signals (CK and CK#)
- Data sampled at crossover point for improved timing precision
- Provides better noise immunity and signal integrity
- Enables tighter timing margins at higher frequencies

Data Strobe (DQS)

DDR introduces bidirectional **DQS (Data Strobe)** signals:

Read Operations:

- DQS generated by memory chip
- Edge-aligned with data output (DQ)
- Controller captures data on DQS edges
- Time from CAS command to DQS = CAS latency (CL)

Write Operations:

- DQS generated by controller
- Center-aligned with DQ for stable sampling window
- Write latency (tDQSS) = delay from write command to first DQS edge
- Typically ~1 clock cycle

DLL (Delay-Locked Loop)

DDR incorporates DLL circuitry to:

- Synchronize internal and external clocks dynamically
- Compensate for voltage and temperature variations
- Maintain timing alignment across clock domains
- Support higher operating frequencies

MEMORY CELL STRUCTURE

DRAM Cell Fundamentals

Each memory cell = **1 transistor + 1 capacitor** (1T1C):

- Capacitor stores charge (bit value: 0V = '0', 1-2V = '1')
- Transistor controls access to capacitor
- Volatile: loses data without power and periodic refresh

Bank Organization

DDR1 Structure:

- Typically 4 independent banks per chip
- Each bank: Rows × Columns architecture
- Bank operations can be interleaved for performance
- Only one row active per bank at a time

Memory Access Phases:

1. **Precharge** (~15-20ns): Prepare bitlines
2. **Row Activation** (~50ns): Select and sense row, latch in sense amplifiers
3. **Column Access** (<10ns): Read/write specific columns from active row

Key Performance Insight: Row activation is slow, but subsequent column accesses to the same open row are fast. Prefetch architecture exploits this by grabbing multiple sequential columns.

TIMING PARAMETERS

Critical Latencies

CAS Latency (CL):

- Clock cycles from READ command to data output
- DDR allows fractional values (2.5, 3, 3.5 cycles possible)
- Fractional due to double-edge operation

tRCD (RAS to CAS Delay):

- Time from row activation to column access command
- Typical: 15-20ns

tRP (Row Precharge Time):

- Time to close active row and precharge
- Must complete before new row activation

tRAS (Row Active Time):

- Minimum time row must remain active
- Ensures data integrity

tRFC (Refresh Cycle Time):

- Time required for complete refresh operation
- ~60-90ns for DDR1

tWR (Write Recovery Time):

- Time from last data written to precharge command
- ~3 clock cycles for DDR

SIGNAL INTERFACE

SSTL_2 (Stub Series Terminated Logic)

DDR uses low-voltage differential signaling:

- **Operating voltage:** 2.5V (vs 3.3V for SDRAM)
- **Reference voltage (VREF):** 1.25V (½ of VDDQ)
- **Termination voltage (VTT):** Also 1.25V
- Logic levels determined relative to VREF (±200-300mV)

Benefits:

- Lower power consumption
- Reduced EMI and crosstalk
- Better signal integrity at higher speeds
- Improved noise margins

Data Mask (DM)

- Replaces SDRAM's output enable for writes
- Masks individual bytes during write operations
- Active HIGH during write to prevent writing invalid data
- No read masking (controller decides what to read)

MODULE ARCHITECTURE

DIMM Configuration (Desktop DDR1)

- **184 pins** (vs 168 for SDRAM, 240 for DDR2)
- **Single notch** for keying (SDRAM has two)
- **64-bit data bus** (8 bytes per transfer)
- Typical: 8 chips of 8-bit width each

Memory Ranks

- **Rank** = group of chips sharing address/command bus
- All chips in rank accessed simultaneously
- Chip select (CS) activates specific rank
- Module can have 1, 2, or 4 ranks (not related to sides)

Example Configurations (1GB module):

- 36 chips × 256Mbit (×4) = dual-rank ECC
- 18 chips × 512Mbit (×8) = dual-rank non-ECC
- 18 chips × 512Mbit (×4) = single-rank

Chip Organization Notation

64M×8 means:

- 64 million addressable locations
- Each location stores 8 bits
- Total capacity: 512 Mbit (64M × 8)

Impact on Features:

- ×4 chips: Enable advanced ECC (Chipkill)
- ×8 chips: Lower cost, common in desktops
- ×16 chips: Fewer chips needed but less flexible

POWER MANAGEMENT

Voltage Specifications

- **DDR1:** 2.5V / 2.6V (overclocked variants)
- **DDR2:** 1.8V
- **DDR3:** 1.5V / 1.35V (DDR3L)
- **DDR4:** 1.2V
- **DDR5:** 1.1V

Power Consumption (DDR1):

- Idle: ~1W per 512MB module
- Active: ~2-3W per 512MB module
- Scales with clock rate and data activity

Refresh Requirements

DDR requires periodic refresh to maintain data:

- **Refresh interval:** Every 64ms typically
- Each refresh addresses one row per bank
- All 4 banks refreshed in rotation
- Auto-refresh or self-refresh modes available
- Refresh commands consume bandwidth and power

DDR GENERATIONS EVOLUTION

Prefetch Architecture Progression

DDR1 (2n prefetch):

- Internal bus = 2× I/O bus
- 2 words per core access
- Supports 100-200 MHz clock

DDR2 (4n prefetch):

- Internal bus = 4× I/O bus
- 4 words per core access
- Same core speed as DDR1, but 2× external speed
- Higher latencies initially due to deeper pipeline

DDR3 (8n prefetch):

- Internal bus = 8× I/O bus
- 8 words per core access
- 4× faster I/O than core
- Example: 200 MHz core + 1600 MT/s I/O

DDR4 (8n prefetch + bank groups):

- Still 8n prefetch (didn't go to 16n)
- **Innovation:** 4 bank groups with 4 banks each
- Each bank group operates semi-independently
- Time-division multiplexing between groups
- Allows sustained high I/O speed without 16n overhead

DDR5 (16n prefetch + bank groups):

- 16n prefetch architecture
- Same-bank group access latency improvements
- Enhanced bank group architecture

Performance Evolution

Generation	Year	Voltage	Prefetch	Clock (MHz)	Data Rate (MT/s)	Bandwidth (GB/s)	Pins
DDR1	2000	2.5V	2n	100-200	200-400	1.6-3.2	184
DDR2	2003	1.8V	4n	100-266	400-1066	3.2-8.5	240
DDR3	2007	1.5V	8n	100-266	800-2133	6.4-17	240
DDR4	2014	1.2V	8n	133-400	1600-3200	12.8-25.6	288
DDR5	2020	1.1V	16n	200-450	3200-6400	25.6-51.2	288

Key Trends:

- Voltage reduced each generation (power efficiency)
- Prefetch depth increased (more parallelism)
- I/O speed increased dramatically
- Core speed remained relatively constant
- Gap between core and I/O bridged by deeper prefetch

DIFFERENCES FROM OTHER RAM TYPES

vs SDRAM (SDR)

Aspect	SDRAM	DDR SDRAM
Data transfer	Rising edge only	Both edges
Prefetch	1n (minimal)	2n
Clock	Single-ended	Differential (CK/CK#)
Voltage	3.3V	2.5V
Data strobe	None	DQS signal
Bandwidth	Base	2× SDRAM
Pins (DIMM)	168	184
CAS latency	Integer only	Fractional possible

Core Advantage: DDR doubles bandwidth without requiring faster core or higher power.

vs Asynchronous DRAM

Aspect	Async DRAM	DDR SDRAM
Synchronization	No system clock	Synchronized to system clock
Access method	Asynchronous control	Clock-edge triggered commands
Performance	Slow, unpredictable timing	Fast, deterministic timing
Interleaving	Limited	Bank interleaving optimized
Modern use	Obsolete	Current standard

vs SRAM

Aspect	SRAM	DDR SDRAM
Cell structure	6 transistors (6T)	1 transistor + 1 capacitor (1T1C)
Refresh needed	No	Yes (every 64ms)
Speed	Very fast (1-2ns)	Slower (50ns+ for row access)
Density	Low (larger cells)	High (smaller cells)
Cost	Expensive	Economical
Use case	CPU cache	Main system memory
Volatility	Volatile	Volatile

vs RDRAM (Rambus DRAM)

Aspect	RDRAM	DDR SDRAM
Architecture	Proprietary serial bus	Industry standard parallel bus
Licensing	Expensive royalties	Open JEDEC standard
Heat	Higher heat output	Lower power consumption
Cost	Very expensive	Economical
Bandwidth	High potential	Competitive in later generations
Market outcome	Abandoned	Dominant standard

Why DDR won: Open standard, lower cost, industry-wide support, backward design compatibility.

TECHNICAL ADVANTAGES OF DDR

1. Bandwidth Doubling Without Frequency Increase

- Exploits both clock edges for 2× throughput
- Avoids thermal and electrical issues of higher frequencies
- Core speed remains manageable

2. Lower Power Consumption

- Reduced voltage signaling (SSTL_2)
- Less switching power than equivalent SDR at same bandwidth
- ~40-50% power reduction vs SDRAM for same performance

3. Prefetch Efficiency

- Matches sequential access patterns of most applications
- Leverages fast column access after slow row activation
- Improves effective bandwidth utilization

4. Bank Interleaving

- Multiple independent banks hide latency
- While one bank activates row, another services data
- Maximizes memory controller efficiency

5. Signal Integrity

- Differential clock reduces jitter
- Data strobe (DQS) ensures robust data capture
- Better noise immunity than single-ended designs

6. Standardization (JEDEC)

- Industry-wide compatibility
- No licensing fees (unlike RDRAM)
- Multiple suppliers compete on price/quality

TECHNICAL LIMITATIONS & CHALLENGES

1. Latency Penalty

- Deeper prefetch pipeline adds latency
- Row activation still ~50ns bottleneck
- Random access patterns suffer vs sequential

2. Minimum Burst Length

- Prefetch architecture enforces minimum burst
- DDR3 (8n) wastes bandwidth on <8-word accesses
- Inefficient for small, scattered reads

3. Refresh Overhead

- Periodic refresh consumes ~5% of bandwidth
- Increases with density (more rows to refresh)
- Power cost during idle periods

4. Temperature Sensitivity

- DRAM cells lose charge faster when hot

- May require faster refresh rates (temp-compensated)
- Timing margins degrade with temperature

5. Complexity of High-Speed Operation

- DLL calibration required
- Careful PCB trace matching critical
- On-die termination (ODT) needed for signal integrity
- More complex than SDR SDRAM

WHY USE DDR SDRAM?

Primary Applications

1. Main System Memory (Desktop/Laptop):

- Optimal cost per GB
- Sufficient bandwidth for CPU/GPU
- Industry standard interface

2. Server & Data Center:

- ECC variants for reliability
- Registered DIMMs for large capacity
- High bandwidth for multi-core processors

3. Embedded Systems:

- SO-DIMM variants for space constraints
- Lower power options (LPDDR)
- Wide temperature range parts available

4. Graphics Cards (GDDR variants):

- Modified DDR architecture
- Optimized for graphics bandwidth
- Higher clock speeds possible

Technical Justification

When DDR is Optimal:

- Large sequential data transfers (video, databases)
- Moderate latency tolerance acceptable
- Cost-sensitive applications

- Power budget constraints
- Industry standard compatibility required

When Alternatives Considered:

- **SRAM (cache):** Need ultra-low latency, small capacity acceptable
- **HBM (High Bandwidth Memory):** GPU/AI accelerators, extreme bandwidth
- **LPDDR:** Mobile devices, power critical
- **Persistent Memory (Intel Optane):** Large capacity + persistence needed

Economic Factors

- Mature manufacturing (yields high, cost low)
- Competitive market (Samsung, Micron, SK Hynix, etc.)
- Wide availability across speed/capacity grades
- Proven reliability and longevity

COMMAND PROTOCOL

Key DDR Commands

ACTIVATE (ACT):

- Opens a row in specified bank
- Latches row data into sense amplifiers
- Row stays open until precharge

READ:

- Accesses column(s) from active row
- Data appears after CL clock cycles
- Auto-precharge option available

WRITE:

- Writes data to column(s) in active row
- Data latency = tDQSS
- Uses DM signal for byte masking

PRECHARGE (PRE):

- Closes active row in bank
- Required before activating different row
- Can target single bank or all banks

REFRESH (REF):

- Auto-refresh: Controller-initiated
- Self-refresh: Low-power standby mode
- Refreshes one row per bank per command

NOP (No Operation):

- Placeholder command
- Used for timing alignment

Mode Register Set (MRS):

- Programs DDR configuration
- Sets CAS latency, burst length, etc.
- Extended mode register (EMRS) for additional settings

Burst Operation

DDR supports programmable burst lengths:

- **BL = 2, 4, 8** (standard)
- Sequential or interleaved addressing
- Auto-precharge option at burst end

Example Read Burst (BL=4, CL=3):

```
Cycle:  0    1    2    3    4    5    6
CMD:    ACT  NOP  NOP  READ NOP  NOP  NOP
Data:   ---  ---  ---  ---  D0   D1   D2/D3
        (Row activate)  (CAS)  (CL=3 wait)  (4 words)
```

ADDRESSING & CAPACITY

Address Multiplexing

DDR uses time-multiplexed addressing:

- **Row address** sent during ACTIVATE
- **Column address** sent during READ/WRITE
- Reduces pin count significantly

Example DDR3 1Gb chip (128M×8):

- 13 row address bits → 8192 rows
- 10 column bits → 1024 columns
- 3 bank address bits → 8 banks

- Total: 8192 × 1024 × 8 × 8 bits = 1 Gbit

Capacity Calculation

Module capacity = Chip capacity × Number of chips ÷ 8 (if non-ECC)

Example:

- 8 chips × 512 Mbit each = 4096 Mbit = 512 MB module
- 16 chips × 512 Mbit each = 8192 Mbit = 1024 MB = 1 GB module

ECC Modules:

- Add 1 extra chip per 8 data chips (9 chips total)
- Extra bits for error detection/correction
- Effective capacity = Total / 9 × 8

PRACTICAL IMPLEMENTATION CONSIDERATIONS

PCB Design Requirements

- **Trace matching:** DQ lines within ±25ps (DDR1)
- **Impedance control:** 50-60Ω typical
- **Layer stackup:** Dedicated ground/power planes
- **Decoupling:** Multiple capacitor values near DRAM

Signal Integrity

- **On-die termination (ODT):** Reduces reflections
- **DQS centering:** Write data capture window optimization
- **Fly-by topology:** Later generations for better signal quality
- **VREF tracking:** Maintains reference across voltage variation

Controller Requirements

- DLL initialization and calibration
- Refresh management
- Bank interleaving logic
- Command scheduling for optimal throughput
- Thermal monitoring (optional)

CONCLUSION

DDR SDRAM achieves doubled data rates through:

1. **Dual-edge clocking** (rising + falling)
2. **Prefetch architecture** (2n for DDR1, scaling up)
3. **Differential clocking** (CK/CK#) for precision
4. **Data strobe (DQS)** for robust capture
5. **Bank organization** enabling parallelism

Its technical superiority over SDRAM stems from bandwidth doubling without proportional frequency/power increase, while its dominance over alternatives like RDRAM comes from open standardization, cost effectiveness, and industry support.

DDR's evolution (DDR2→DDR3→DDR4→DDR5) continues the core principles while adding deeper prefetch, bank groups, and voltage reduction to meet ever-increasing performance demands of modern computing.

END OF REPORT